Assignment-2

Group-036

Task 1 - Slate Health Check

This is a high-level overview of the financial health and composition of the studios historical slate.

Some cleaning has been done to prepare the data for this report. Some films in the dataset had missing or unrealistically low values for budget or revenue. Most dollar amounts in the dataset have been entered as the whole figure, but in some cases it appears they have been recorded as hundreds of thousands or millions of dollars. For example, *Escape from Tomorrow* has a recorded budget of 650 in the dataset. The film's actual reported budget was $650,000.

Given the large initial size of the dataset (no. of films = 4803) we decided to filter out any films with budget or revenue amounts less than 1000 to cut films with missing or implausible values. We also removed any films with a reported budget higher than 550 million USD (the largest budget ever reported for a film). The franchise tag is not used for any film pre-1974 even when films *were* part of a franchise (like James Bond); given the importance of comparison between Original and Franchise films to this report we have also filtered for only films released in 1974 or later.

This left us with 3060 films to analyse, of which franchise films made up 7.22% of the films in our cleaned dataset, vs the 5.41% they made up in the original dataset. We are happy that this does not represent a large enough change in the composition of our data to skew our analysis.

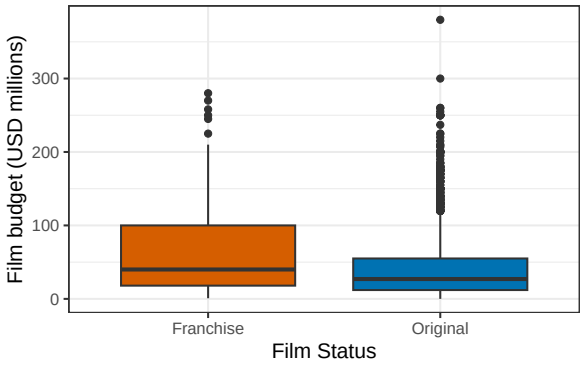
Summary of data

> **Note 1: Coding for Original vs Franchise**
>
> First films in a franchise are labelled as "Original" rather than as part of a franchise, and are therefore included in the data for the "Original" variable (see Table 1, where *Paranormal Activity* is an Original film, and *Paranormal Activity 2* is a Franchise film.)
> Additionally, some films that are clearly part of a franchise are not labelled as such; i.e. all 4 Pirates of the Caribbean movies are labelled as Original movies.
> As such, the figures in Table 1 should be taken as *indicative* of overall trends only, not treated as absolute truths.

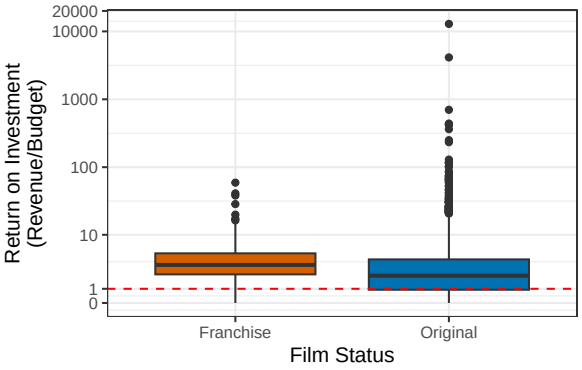
Table 1: Summary statistics for the historical slate

Statistic	Franchise	Original
Number of movies	221	2839
Most Common Genre	Action (n = 63)	Drama (n = 681)
Mean Production Budget (USD millions)	65.15	40.88
Median Production Budget (USD millions)	40	27
Production Budget Standard Deviation (USD millions)	63.2	42.48
Mean Return on Investment (ROI)	4.84	11.05
Median ROI	3.36	2.13
ROI Standard Deviation	6.11	254.93
Max Production Budget (USD millions)	280 *(Avengers: Age of Ultron)*	380 *(Pirates of the Caribbean: On Stranger Tides)*
Minimum Production Budget (USD)	852,510 *(4 Months, 3 Weeks and 2 Days)*	7,000 *(Primer)*
Best ROI	59.17 *(Paranormal Activity 2)*	12890.39 *(Paranormal Activity)*
Worst ROI	0 *(5 Days of War)*	0 *(The Adventurer: The Curse of the Midas Box)*

Budget and ROI plots



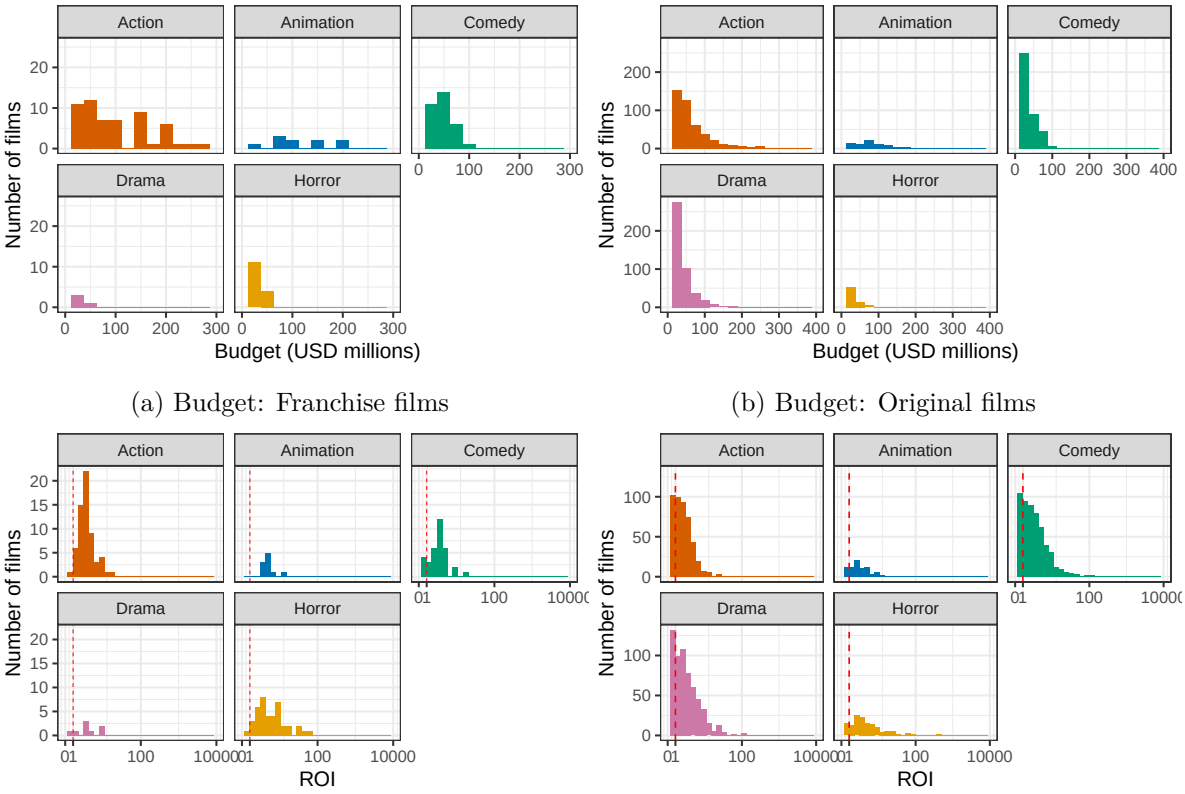
(a) Budget



(b) ROI; dashed red line is at ROI = 1 i.e. films above the line made a profit, films below lost money

Figure 1: Boxplots showing distribution of budgets and Return on Investment (Revenue divided by Budget) for Original vs Franchise films

We already knew from Table 1 what genres appeared the most for Franchise and Original movies. Figure 2 shows that while Drama is the most popular genre for Original films, Action isn't far behind in terms of the number of movies made. Figure 2b also might explain *why* Drama is such a popular genre for Original movies; there's a lot of low-budget movies Drama movies being made! The same budget pattern can be seen for Original Comedy films, although interestingly they seem to differ in terms of ROI pattern. It seems more likely that you'll make your money back on an original comedy than on an original drama. The same is not true for franchise comedies; comedy sequels seem to be the riskiest investment, with a much higher percentage of films falling below the magic ROI equals one line than for any other Franchise



(a) Budget: Franchise films

(b) Budget: Original films

(c) ROI: Franchise films

(d) ROI: Original films

Figure 2: Histograms showing distribution of budget costs and ROI for Original and Franchise films, faceted by film genre; only Meridian's 5 focused genres are included

genre.

Slate composition over time

We wante to look at how the composition of the historical slate by genre has changed over time, with a specific focus on the genres Meridian is interested in.

Worth noting is that although we have some data from 2015 on, it is not complete data and is therefore not included in **?@fig-genre-line-chart**. It is unclear if the data we have from 2014 is fully complete, but it has been included until we are given a compelling reason to exclude it.

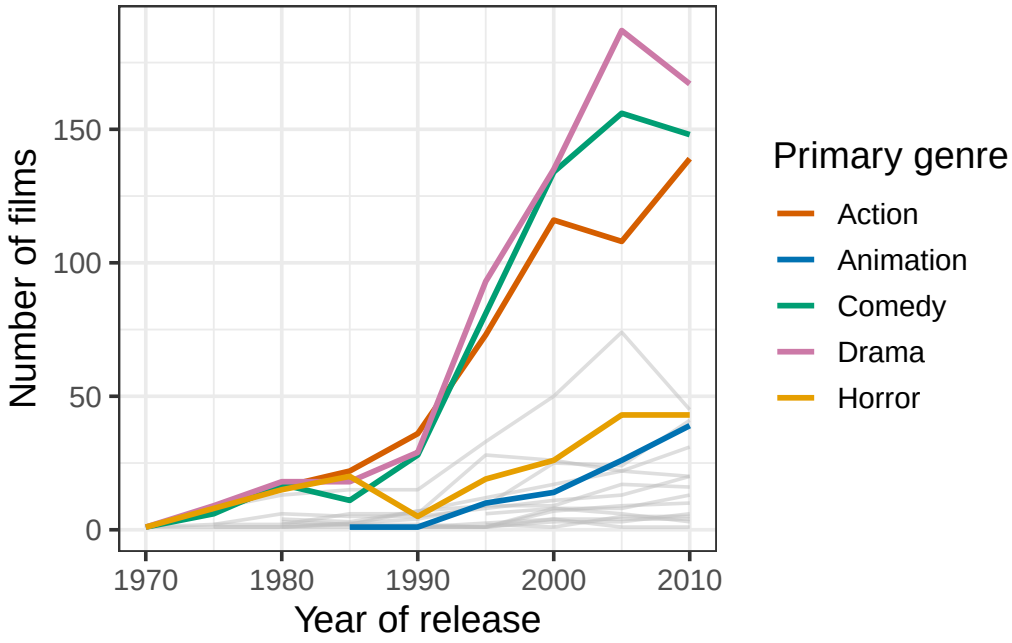


Figure 3: Line chart showing trend in genre releases over time; the 5 genres Meridian is focused on are highlighted; film releases have been aggregated into 5 year blocks to make trends more visible

Summary

From the plots in Figure 1, and given what we know about how films are coded (Note 1), we can draw some conclusions from the above plots:

1. Sequels are generally more expensive to make than original films

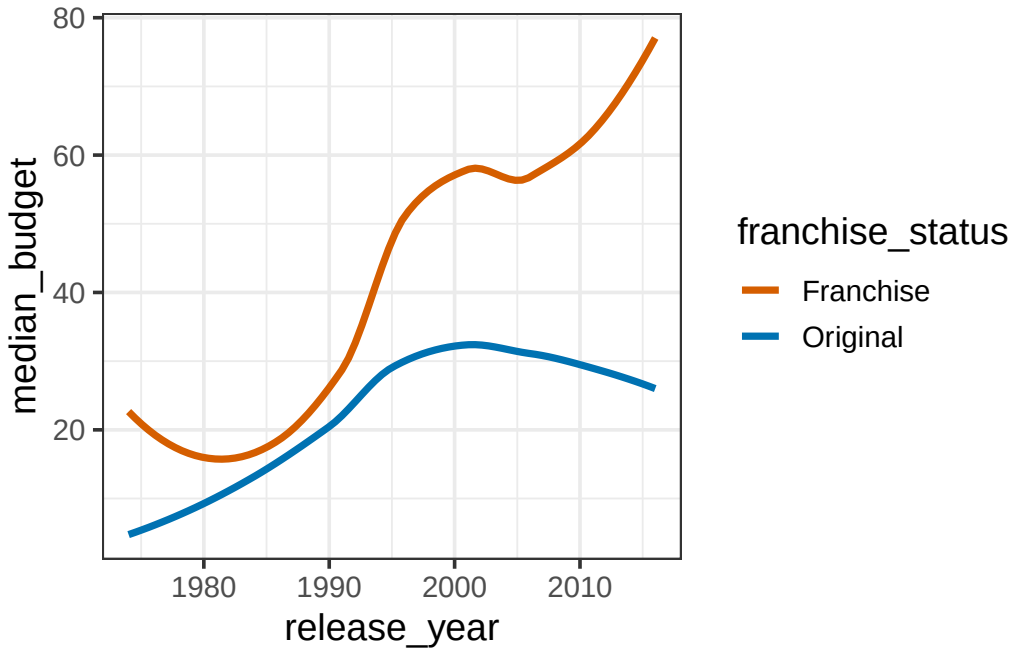
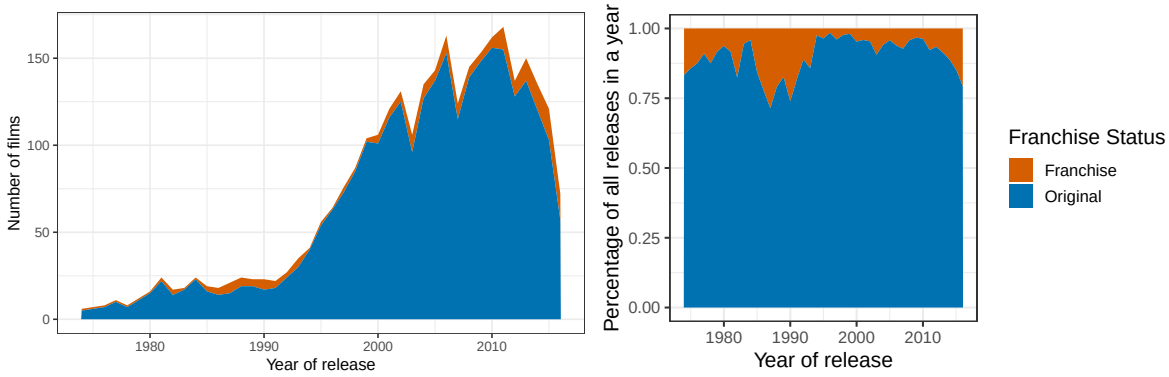


Figure 4: Line chart showing trend in genre releases over time; the 5 genres Meridian is focused on are highlighted; film releases have been aggregated into 5 year blocks to make trends more visible



(a) Stacked area chart displaying actual numbers of films for each year

(b) Filled area chart showing franchise status as a percentage of all films released in a given year

Figure 5: Area charts showing the split between original and franchise films by year

2. Sequels are generally a safer bet than original films; they have a higher ROI on average, and are less likely to fall below an ROI of 1 (meaning they generally at least make their money back)
3. Original films have a higher ceiling for ROI
    - This is probably at least somewhat a function of point 1
    - Original movies are often made for smaller amounts, and if an original movie is a surprise hit and does reasonably well at the box office on a small budget, ROI will look enormous

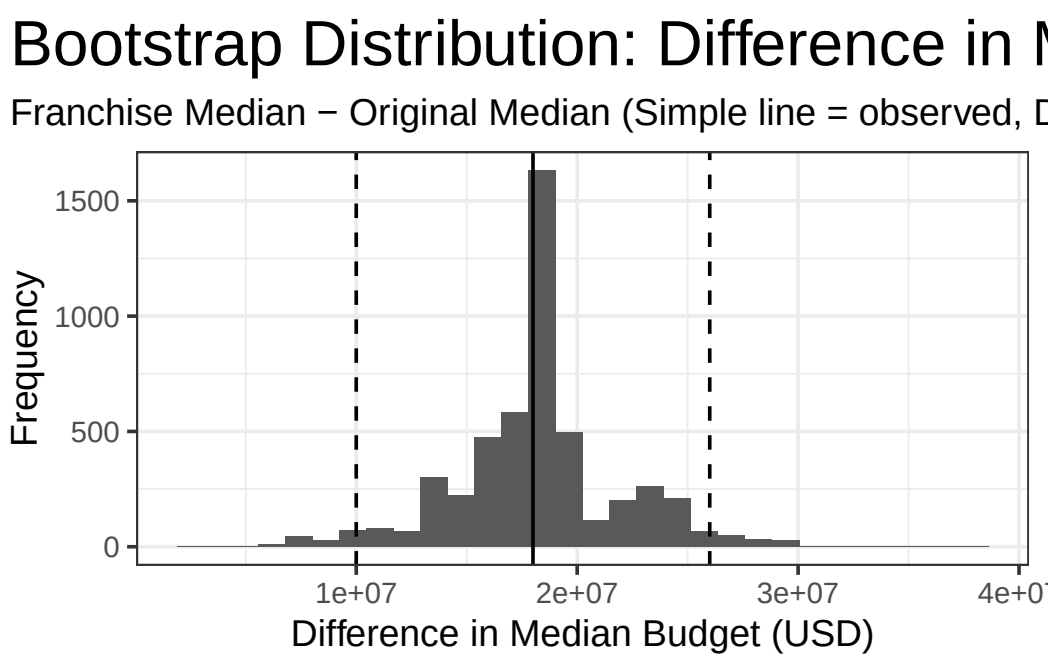
Task 2

Budget data are often skewed, and the median is often more meaningful than the mean when describing a budget for typical franchise and original categories. Here I will be using bootstraping to find 95% interval estimate for the difference in median budget between the two groups. As bootstraping simulates repeated samples from a population like the csv provided here, it can help us to describe the uncertainity in the data.

As interpreted from the Figure 1 (a) and the Table **??**, the 'original' franchise status have the lower median which shows that the production budgets are biased towards the 'franchise' category. To confirm this we will use bootstraping. The point estimate of the data is 1.8×10^7 which we will use to confirm if bootstarped estimate is similar or not. If yes, then the budget allocation is truely are leaned towards 'franchise' category.

The percentile bootstrap confidence interval of the data is 10^7 and 2.6×10^7. The interval suggests that the population median difference is plausible between these values based on the data and bootstrap method.

Does it verify leadership concern?



This shows that the median difference is well above zero and similar to the point estimate which supports the concerns of the leadership of franchise slate absorbing much larger budgets than original slate.

Additionally from Section Figure 1, we can interpret that the originals have higher number of outliers that drastically increases the profits of the production houses. However, the overall ROI in origianls is low.

Summary

The median is the best way to summarise the budget data as they are always right skewed and using mean won't provide any acceptable information. We needed to know about the difference in median of the two facets of franchise status to confirm the concern of the leadership about the baised budget allocation towards franchises. By boostraping the median difference of the `franchise_status` categories and creating a histogram of it showed that the median difference is well above zero (near 18 Million) which confirms the concern about budget allocations.

However, ss stated in Section , the sequels are much expensive to make than the originals and they have higher ROI than the originals, due to this reason the production houses are more inclined towards producing sequals as they have much better chance of creating profits than the originals. But from Section Figure 1, we can interpret that the originals have higher number

of outliers that drastically increases the profits of the production houses. However, the overall ROI in origianls is low.

Advice

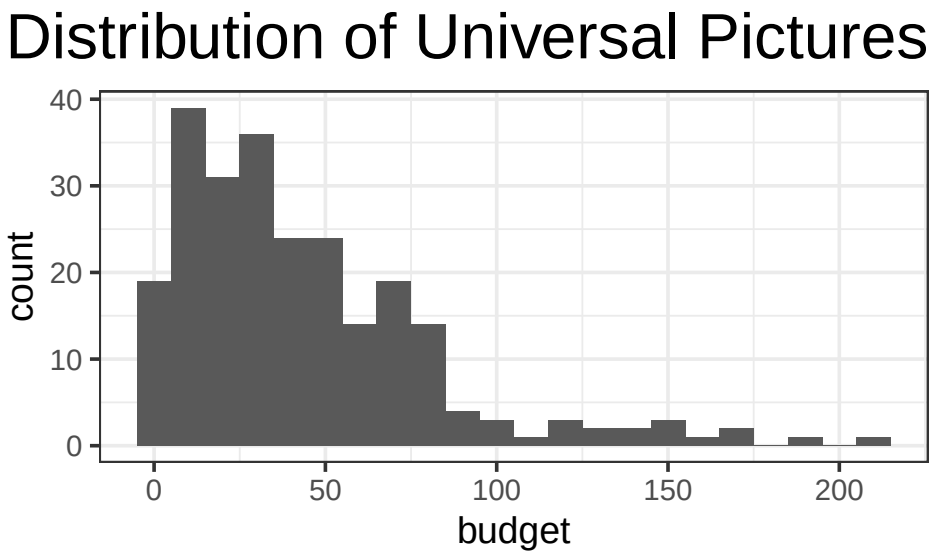
The advice to the leadership is to focus on original movies that have high potential such as the genre that have the most profitable originals and gradually increase the budget allocation for them and compare the profits generated by both categories simultaniously and changing or refining the new original story selection strategy, find a optimised allocation stratergy that maximises the profits for the company.

Choosing more originals carefully and strategically, provides more room for higher profits and gradually trying and testing original category in different genre and creating an optimal budget allocation for movies can increase the profits with a high margin.

Task 3

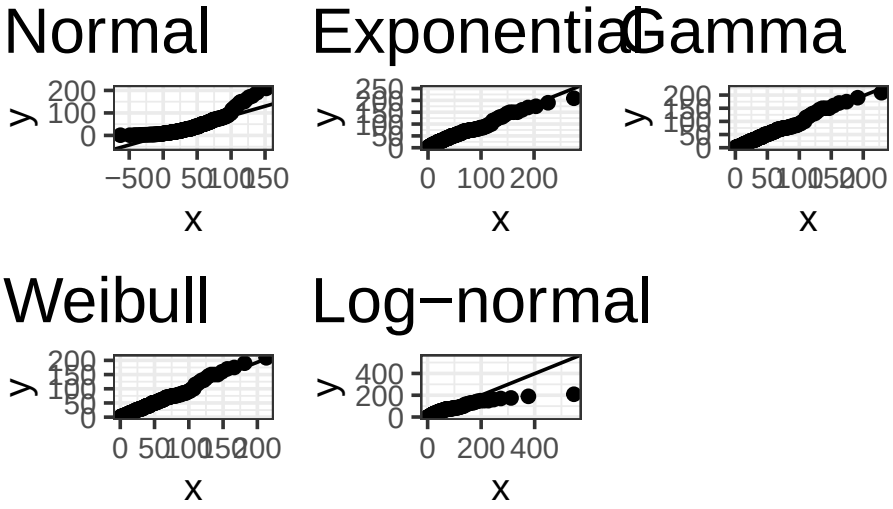
Part 1: Production Budget Modelling

With the goal to construct a model that models an appropriate movie budget distribution, a competitor budget analysis was conducted as a credible reference point in line with industry standards. This was performed by sampling the budgeting model from Universal Pictures Studios, which is a direct industry competitor, and fitting a realistic probability model for Meridian to follow.



model	df	AIC
budget_gamma	2	2320
budget_weibull	2	2321
budget_exp	1	2332
budget_lognormal	2	2349
budget_normal	2	2454

After data processing, the budget distribution is right skewed and only contain positive values. Therefore, exponential, Gamma, Log normal and Weibull distribution can be selected for model evaluation.



The qq plot indicates normal and log normal are the worst fits as tails deviate the most from the reference line. The Exponential and Gamma model have the best fist as the right tail have the smallest distance from the reference line and most of the points overlap with the line.

The Gamma model has lowest AIC, which paired with the fitting of the qqplot, which suggests that it is the best at fitting budget data.

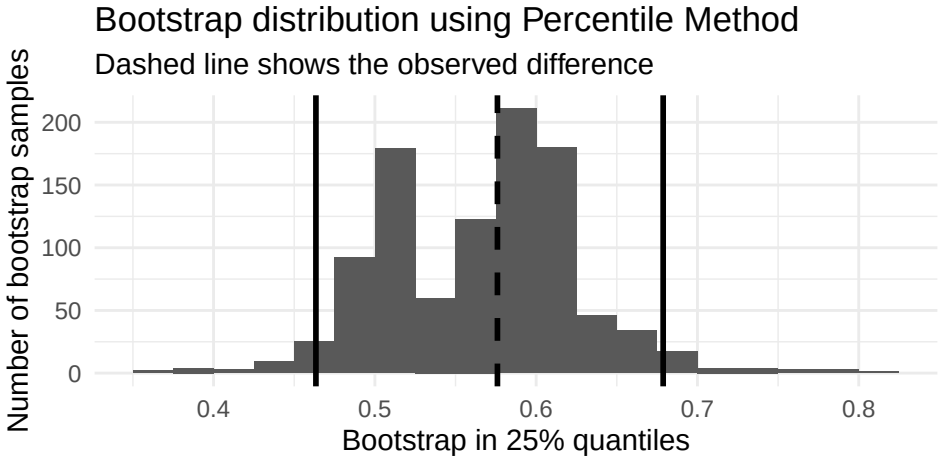
Part 2: Downside Risk Analysis for Action Movie ROI

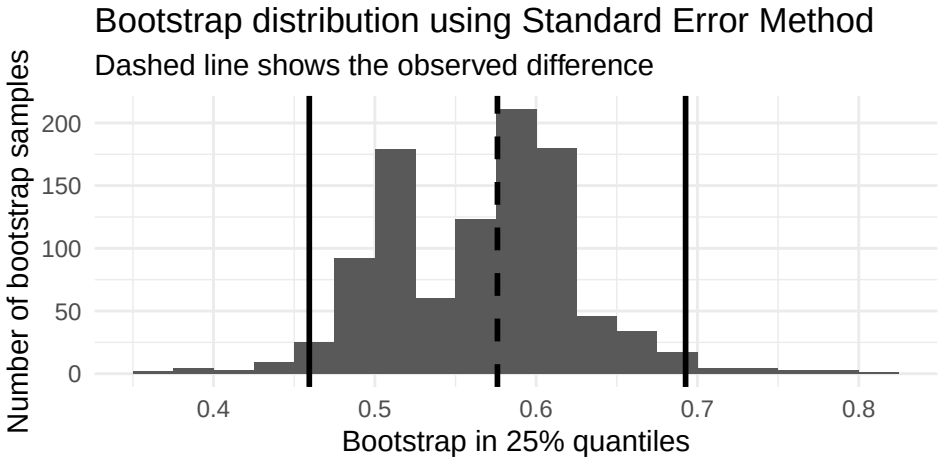
Action movies have been established as capital-intensive to produce, which can be proved as risky for budget investment in order to yield significant revenue. As a result, risk analysis into

Calculation_method	Lower_Bound	Upper_bound
Percentile bootstrap	0.46	0.68
Bootstrap SE (t)	0.46	0.69

the bottom performing 25% movie ROI was performed across the entire film industry to model conservative target ROI to aim for,

The observed 25% percentile of Action Movie ROI was 0.58





Both methods to calculate the confidence interval use the bootstrapped sample, which assumes the sample of Action films is representative of the population. Overall, the results were

method	gap_below	gap_above	width
Percentile bootstrap	0.11	0.10	0.22
Bootstrap SE (t)	0.12	0.12	0.23

similar with the percentile interval (0.463, 0.679) and the standard error interval (0.460, 0.693) overlapping significantly.

The percentile method does not consider symmetry as it takes the 2.5th and 97.5th percentiles directly from the bootstrapped distribution, while the bootstrapped standard error method assumes the distribution is symmetric and approximately bell-shaped which places the lower and upper bounds equidistant from the observed 25th percentile (0.576). The Standard Error method slightly overstated the upper bound in comparison to the percentile bootstrap estimate while maintaining similar estimates for the lower bound. This may have been because the bootstrapped distribution is slightly left skewed where the percentile bootstrap method displays a larger gap between the observed estimate and the lower bound than the upper bound. The Standard error method also had a larger width in the confidence interval likely due to the larger tails, which increase the estimated Standard Error as a result. Since the percentile method does not rely on the symmetry assumption being accurate, it is best to use this method although both methods yielded similar results.

Summary

Based on the analysis, it is recommended to select the Gamma distribution for the simulation. It had the best fit for the budget data as it matched the QQ-plot reference line the closest and had the lowest AIC value after evaluating five different models. It captured the right skew in budgets where few movie budgets had more expensive budgets compared to the majority of movies.

In this sample, 25% of action films had an ROI of 0.576 or lower. We are 95% confident the true 25th percentile of the ROI for all action films across all studios lies between 0.463 and 0.679. For a conservative risk figure to plan around, Finance should consider 0.463 (46.3% return) for an underperforming Action film.

Task 4

```
Rows: 3,060
Columns: 8
$ title            <chr> "Avatar", "Pirates of the Caribbe~
$ budget           <int> 237000000, 300000000, 245000000, ~
$ roi              <dbl> 11.76, 3.20, 3.59, 4.34, 1.09, 3.~
$ franchise_status <fct> Original, Original, Franchise, Or~
```

```
$ primary_genre    <chr> "Action", "Adventure", "Action", ~
$ release_year     <dbl> 2009, 2007, 2015, 2012, 2012, 200~
$ cast_size        <int> 83, 34, 83, 158, 27, 143, 13, 72,~
$ flop             <fct> Not flop, Not flop, Not flop, Not~
```

The new flop variable only has two possible outcomes, which means it can be modeled by a Bernoulli distribution, where the parameter p measures the probability that a move is a flop. Hence, logistic regression is the appropriate choice as the log function $\log\left(\frac{p_i}{1-p_i}\right)$ ensures the predicted probabilities remain between 0 and 1. The linear regression is not suitable as it assumes outcomes are continuous and normally distributed, while the flop variable only takes 0 and 1. Poisson regression is also not suitable as it is used when the outcome is a count with no upper limit.

```
Call:
glm(formula = flop ~ primary_genre + franchise_status + log_budget.z +
    cast_size.z, family = binomial, data = task4_data)

Coefficients:
                           Estimate Std. Error z value
(Intercept)                 10.3467   324.7438    0.03
primary_genreAction        -12.7217   324.7437   -0.04
primary_genreAdventure     -13.1035   324.7438   -0.04
primary_genreAnimation     -13.4116   324.7438   -0.04
primary_genreComedy        -13.0635   324.7437   -0.04
primary_genreCrime         -12.9602   324.7438   -0.04
primary_genreDocumentary   -13.0673   324.7440   -0.04
primary_genreDrama         -12.7410   324.7437   -0.04
primary_genreFamily        -13.1324   324.7440   -0.04
primary_genreFantasy       -13.5623   324.7438   -0.04
primary_genreForeign        -0.0183   459.2569    0.00
primary_genreHistory       -12.9150   324.7444   -0.04
primary_genreHorror        -13.8445   324.7438   -0.04
primary_genreMusic         -12.8481   324.7442   -0.04
primary_genreMystery       -13.1917   324.7441   -0.04
primary_genreRomance       -13.4425   324.7439   -0.04
primary_genreScience Fiction -13.3285   324.7439   -0.04
primary_genreThriller      -13.1633   324.7438   -0.04
primary_genreWar           -11.4384   324.7442   -0.04
primary_genreWestern       -13.3199   324.7447   -0.04
franchise_statusOriginal     1.5449     0.2945    5.25
log_budget.z                -0.0270     0.0478   -0.57
```

```
cast_size.z                  -0.5502     0.0688   -8.00
                            Pr(>|z|)
(Intercept)                     0.97
primary_genreAction             0.97
primary_genreAdventure          0.97
primary_genreAnimation          0.97
primary_genreComedy             0.97
primary_genreCrime              0.97
primary_genreDocumentary        0.97
primary_genreDrama              0.97
primary_genreFamily             0.97
primary_genreFantasy            0.97
primary_genreForeign            1.00
primary_genreHistory            0.97
primary_genreHorror             0.97
primary_genreMusic              0.97
primary_genreMystery            0.97
primary_genreRomance            0.97
primary_genreScience Fiction    0.97
primary_genreThriller           0.97
primary_genreWar                0.97
primary_genreWestern            0.97
franchise_statusOriginal     1.6e-07 ***
log_budget.z                    0.57
cast_size.z                  1.2e-15 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3446.9  on 3057  degrees of freedom
Residual deviance: 3241.4  on 3035  degrees of freedom
AIC: 3287

Number of Fisher Scoring iterations: 11

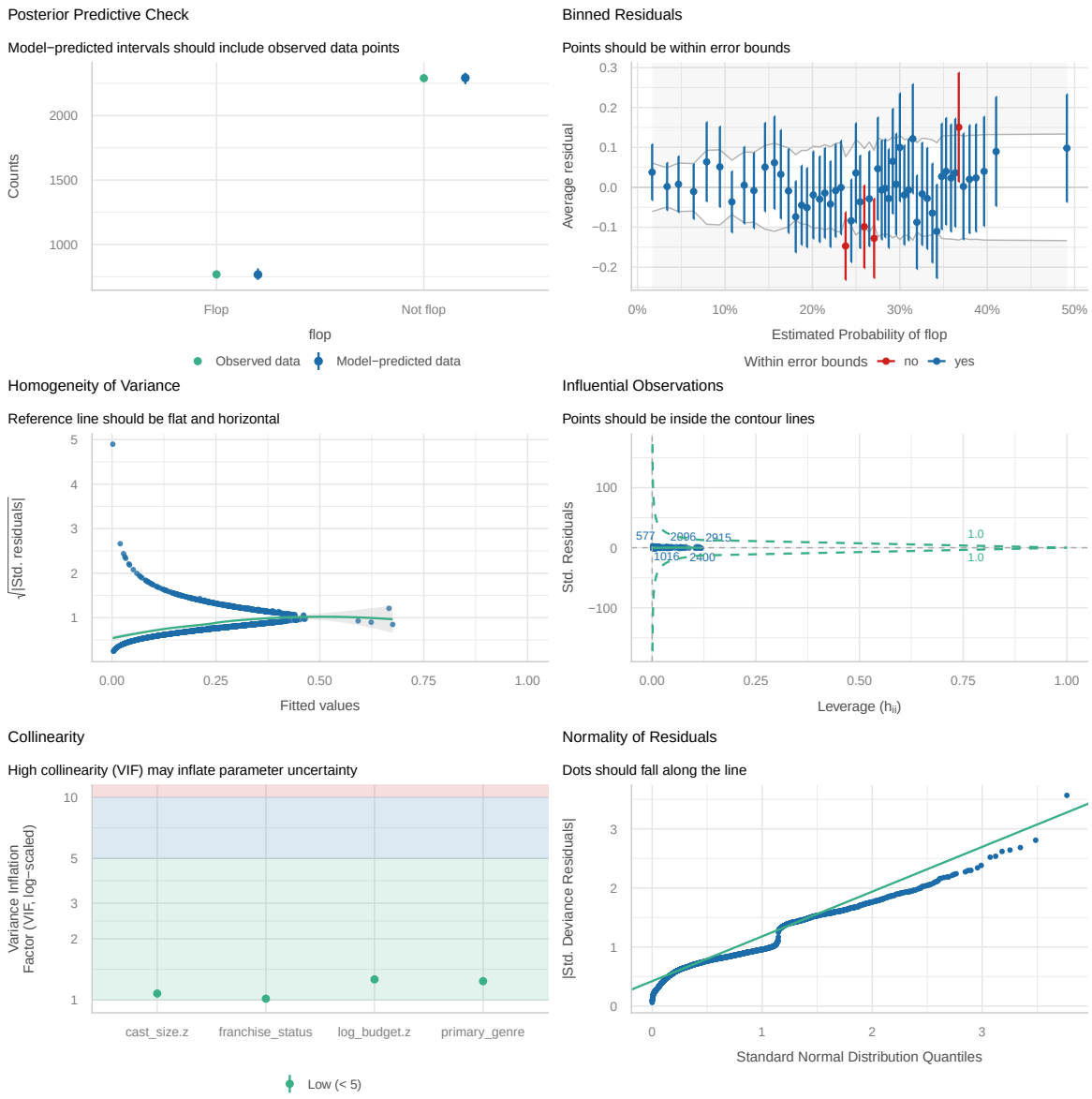

                             2.5 % 97.5 %
(Intercept)                 -55.81     NA
primary_genreAction             NA 53.459
primary_genreAdventure          NA 53.073
primary_genreAnimation          NA 52.753
```

```
primary_genreComedy              NA 53.119
primary_genreCrime               NA 53.215
primary_genreDocumentary         NA 53.078
primary_genreDrama               NA 53.442
primary_genreFamily              NA 53.010
primary_genreFantasy             NA 52.600
primary_genreForeign          -1.71  1.645
primary_genreHistory             NA 53.163
primary_genreHorror              NA 52.329
primary_genreMusic               NA 53.263
primary_genreMystery             NA 52.939
primary_genreRomance             NA 52.718
primary_genreScience Fiction     NA 52.831
primary_genreThriller            NA 53.009
primary_genreWar                 NA 54.666
primary_genreWestern             NA 52.728
franchise_statusOriginal       1.01  2.171
log_budget.z                  -0.12  0.067
cast_size.z                   -0.69 -0.419
```

```
                (Intercept)          primary_genreAction
                    3.1e+04                      3.0e-06
      primary_genreAdventure       primary_genreAnimation
                    2.0e-06                      1.5e-06
         primary_genreComedy           primary_genreCrime
                    2.1e-06                      2.4e-06
    primary_genreDocumentary           primary_genreDrama
                    2.1e-06                      2.9e-06
         primary_genreFamily         primary_genreFantasy
                    2.0e-06                      1.3e-06
        primary_genreForeign         primary_genreHistory
                    9.8e-01                      2.5e-06
         primary_genreHorror           primary_genreMusic
                    9.7e-07                      2.6e-06
        primary_genreMystery         primary_genreRomance
                    1.9e-06                      1.5e-06
primary_genreScience Fiction        primary_genreThriller
                    1.6e-06                      1.9e-06
             primary_genreWar         primary_genreWestern
                    1.1e-05                      1.6e-06
    franchise_statusOriginal                 log_budget.z
                    4.7e+00                      9.7e-01
```

```
                cast_size.z
                    5.8e-01
```

```
                                2.5 %  97.5 %
(Intercept)                 5.8e-25      NA
primary_genreAction              NA 1.6e+23
primary_genreAdventure           NA 1.1e+23
primary_genreAnimation           NA 8.1e+22
primary_genreComedy              NA 1.2e+23
primary_genreCrime               NA 1.3e+23
primary_genreDocumentary         NA 1.1e+23
primary_genreDrama               NA 1.6e+23
primary_genreFamily              NA 1.1e+23
primary_genreFantasy             NA 7.0e+22
primary_genreForeign        1.8e-01 5.2e+00
primary_genreHistory             NA 1.2e+23
primary_genreHorror              NA 5.3e+22
primary_genreMusic               NA 1.4e+23
primary_genreMystery             NA 9.8e+22
primary_genreRomance             NA 7.9e+22
primary_genreScience Fiction     NA 8.8e+22
primary_genreThriller            NA 1.1e+23
primary_genreWar                 NA 5.5e+23
primary_genreWestern             NA 7.9e+22
franchise_statusOriginal    2.7e+00 8.8e+00
log_budget.z                8.9e-01 1.1e+00
cast_size.z                 5.0e-01 6.6e-01
```



Franchise orginal and cast size had little statistical st

The frequentist model reproduces the observed numbers of flops and non-flops. All VIFs are close to 1, which rules out colinearity. No individual film is overly influential. Most binned residuals fall within the error bounds, although some residuals around predicted probabilities of 20–30% fall were below the error interval. The homogeneity of variance and normality plots are not relevant for a binary outcome. We also assume the films are independent and that each numeric predictor has a linear relationship with the log-odds of a flop.

	Action	Adventure
1	566	270
Animation	Comedy	Crime
96	609	135
Documentary	Drama	Family
27	691	37
Fantasy	Foreign	History
91	1	13
Horror	Music	Mystery
193	17	25
Romance	Science Fiction	Thriller
69	76	116
War	Western	
15	10	